

Module:	User-service					
Github:	https://github.com/trungtoto/CVConnect-Unit-Test					
Total TCs:	33					
Pass:	33					
Fail:	0					
Test Case ID	Test Objective	Input	Expected Output	Tester	Status	Note
UserServiceImplTest						
TC-US-USER-001	Reject resetPassword when user does not exist	userId=100,newPassword	Throw AppException USER_NOT_FOUND and no save	Thái	PASS	CheckDB + Rollback (no DB mutation)
TC-US-USER-002	Reject resetPassword for third-party account	userId=1,accessMethod=GOOGLE	Throw AppException REGISTER_THIRD_PARTY and no save	Thái	PASS	CheckDB + Rollback (no DB mutation)
TC-US-USER-003	Update encoded password for LOCAL account	userId=2,newPassword	Repository save with encoded password	Thái	PASS	CheckDB update path
TC-US-USER-004	Reject updateEmailVerified when user missing	userId=88,emailVerified=true	Throw AppException USER_NOT_FOUND and no save	Thái	PASS	CheckDB + Rollback (no DB mutation)
TC-US-USER-005	Persist new email verification flag	userId=3,emailVerified=false	Repository save with isEmailVerified=false	Thái	PASS	CheckDB update path
TC-US-USER-006	Return empty list when no users by role and org	roleCode=HR,orgId=55	Return empty list	Thái	PASS	DB read empty branch
TC-US-USER-007	Map users by role and hide password in response	roleCode=ORG_ADMIN,orgId=9	Return mapped DTO with password=null	Thái	PASS	DB read + response safety
OrgMemberServiceImplTest						
TC-US-OM-001	Return null when org-member record not found	userId=1	Return null and no org API call	Thái	PASS	DB read not found
TC-US-OM-002	Return null for inactive org-member	userId=2,isActive=false	Return null	Thái	PASS	DB read inactive branch
TC-US-OM-003	Map active org-member and organization	userId=3,orgId=100	Return OrgMemberDto with org info	Thái	PASS	DB read + enrichment
TC-US-OM-004	Block invite when user already in same org	orgId=500,userId=11	Throw AppException USER_JOINED_ORG and no invite created	Thái	PASS	CheckDB + Rollback (no DB mutation)
TC-US-OM-005	Block invite when target email not verified	orgId=600,userId=12,emailVerified=false	Throw AppException EMAIL_NOT_VERIFIED and no invite created	Thái	PASS	CheckDB + Rollback (no DB mutation)
TC-US-OM-006	Create invite and send email on valid request	orgId=700,userId=13,roleId=23	Persist PENDING invite with token and send email	Thái	PASS	CheckDB insert path
JobConfigServiceImplTest						
TC-US-JC-001	Filter active jobs and map DTO fields	active+inactive+null active configs	Only active job returned and mapped	Thái	PASS	Branch coverage for Boolean.TRUE filter
JobSchedulerTest						
TC-US-JS-001	Do not schedule when DB job config is empty	jobConfigs=[]	No TaskScheduler interaction	Thái	PASS	DB read empty branch
TC-US-JS-002	Skip scheduling for unknown job name	jobName=missing-job	No TaskScheduler interaction	Thái	PASS	Registry miss branch
TC-US-JS-003	Schedule CRON job	scheduleType=CRON,expression=0 0/5 * * *	TaskScheduler schedule called with CronTrigger	Thái	PASS	Scheduling CRON path
TC-US-JS-004	Schedule FIXED_RATE job in milliseconds	scheduleType=FIXED_RATE,expression=7	TaskScheduler scheduleAtFixedRate called with 7000	Thái	PASS	Seconds to milliseconds branch
TC-US-JS-005	Schedule FIXED_DELAY job in milliseconds	scheduleType=FIXED_DELAY,expression=9	TaskScheduler scheduleWithFixedDelay called with 9000	Thái	PASS	Seconds to milliseconds branch
FailedRollbackHandlerRegistryTest						
TC-US-FRREG-001	Resolve handler by rollback type	types=ORG_CREATION,UPLOAD_FILE	Return matching handler instance	Thái	PASS	Registry lookup success
TC-US-FRREG-002	Return null for unregistered handler type	registered=ORG_CREATION, request=UPLOAD_FILE	Return null	Thái	PASS	Registry lookup miss
FailedRollbackServiceImplTest						
TC-US-FRS-001	Save FailedRollback DTO to repository entity	type=UPLOAD_FILE,payload,error,status, retry	Repository save called with mapped entity	Thái	PASS	CheckDB write path
TC-US-FRS-002	Return empty list when no pending rollback records	pending=[]	Return empty list	Thái	PASS	DB read empty branch
TC-US-FRS-003	Map pending rollback entities to DTO list	entity id=99,type=ORG_CREATION	Return mapped DTO with expected fields	Thái	PASS	DB read mapping branch
FailedRollbackRetryJobTest						
TC-US-FRJOB-001	Validate retry job metadata	getJobName.getScheduleType.getExpression	Expected constants and empty schedule metadata	Thái	PASS	Metadata contract
TC-US-FRJOB-002	Do nothing when no pending rollback records	pending=[]	No handler lookup and no save	Thái	PASS	DB read empty branch
TC-US-FRJOB-003	Skip invalid rollback type	type=UNKNOWN_TYPE	No handler lookup and no save	Thái	PASS	Invalid enum branch
TC-US-FRJOB-004	Skip when valid type has no handler	type=ORG_CREATION,handler=null	No save	Thái	PASS	Missing handler branch
TC-US-FRJOB-005	Mark rollback success and persist status	type=UPLOAD_FILE,rollback success	status=true and save called	Thái	PASS	CheckDB update path
TC-US-FRJOB-006	Increase retryCount when rollback fails	type=ORG_CREATION,rollback throws exception	retryCount incremented and save called	Thái	PASS	Rollback compensation path
JwtUtilsTest						

TC-US-JWT-001	Generate JWT with expected claims	valid user,roles,permissions,base64 secret	Non-empty JWT with user/roles/permissions claims	Thái	PASS	Happy path token generation
TC-US-JWT-002	Throw standardized error when secret key invalid	SECRET_KEY=null	Throw ApplicationException CommonErrorCode.ERROR	Thái	PASS	Error handling branch
TC-US-JWT-003	Generate non-empty unique helper tokens	refresh,verify,reset,invite methods	All tokens non-empty and refresh tokens unique	Thái	PASS	Utility token generation